

Q4.

To hover at rest,

Upward force on suit = weight of suit = W

By Newton's 3rd Law,

Downward force that accelerate ions = upward force = W

By Newton's 2nd Law,

$$F = \frac{dp}{dt} = \frac{dm}{dt} \times \Delta v$$

$$\frac{dm}{dt} \times 30,000 = 250 \times 9.81$$

$\therefore$ mass of ions expelled per unit time = $dm/dt = 0.082 \text{ kg s}^{-1}$.

Ans: B

Q5